

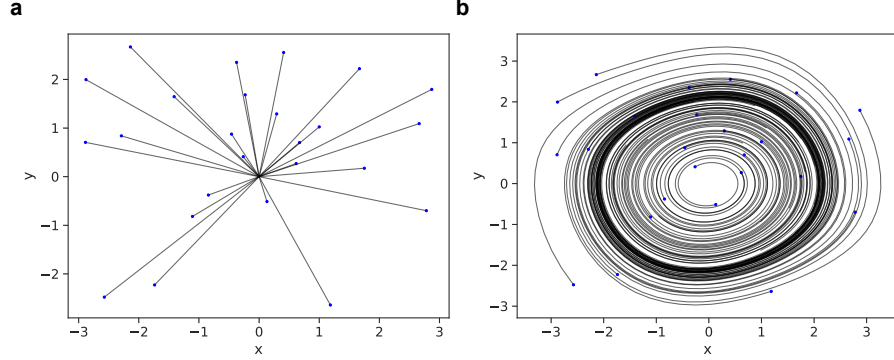


Figure 3: **Trajectories of Attractor Dynamics.** **a, b,** Examples of point and cyclic attractor dynamics, respectively, with 25 random initial positions. These trajectories were generated based on Equations 1 and 2, respectively. The initial positions are denoted by blue dots.

to the fixed point attractor located at $[0, 0]$ with gradual deceleration. Cyclic attractors, on the other hand, were generated based on the van der Pol oscillator, shown in equation 2, when the parameter μ was set to 0.1, and T was set to 200. Figure 3 illustrates examples of trajectories corresponding to point and cyclic attractors, each initiated from 25 random initial values.

$$\begin{cases} \frac{dx}{dt} = ax, \\ \frac{dy}{dt} = -y \end{cases} \quad (1)$$

$$\begin{cases} \frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0, \\ y = \frac{dx}{dt} \end{cases} \quad (2)$$

RNN model. The RNN model comprises two layers: an Elman recurrent layer and an output layer. The Elman recurrent layer consists of 20 hidden units with a tanh activation function. In the training phase, the initial hidden state is initialized as a trainable parameter, allowing it to adapt to each of the different trajectories. In the evaluation phase, it is initialized to a value of 0

[29].

Transformer model. The Transformer model used in this study is a decoder-only sequence model. This is a well-used architecture to generate the next words/actions in the sequence in language/action modeling, such as GPT-based models or Robotic Transformer [30, 31]. The Transformer model comprises four layers: an embedding layer, a positional encoding layer, a Transformer decoder layer, and an output layer. The embedding layer consists of 20 hidden units, and the Transformer decoder layer employs 40 units with four multi-heads attention. For the positional encoding layer (PE), we use the trigonometric functions proposed in [8] formulated as:

$$\begin{cases} PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) \\ PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right), \end{cases} \quad (3)$$

where, pos is the position, i is the dimension, and d_{model} has the same dimensions as the embedding layer. During the training phase, a masked self-attention technique was employed in the decoder to prevent attending to subsequent positions in the sequence.

Training settings. The networks are trained in a supervised manner by minimizing the mean squared error between the network prediction and the next position. The Adam optimizer [32] with $\alpha = 0.001, \beta_1 = 0.9, \beta_2 = 0.999$, and $\epsilon = 10^{-8}$ has been used for the implementation. The networks are trained for 25,000 epochs using NVIDIA GeForce RTX 2080 Ti GPU.

Evaluation settings. We assessed network performance on two types of attractor dynamics by giving 10 random initial positions. Networks were required to autoregressively generate subsequent trajectories based on the given initial values. To quantitatively evaluate the quality of trajectory generation, we adopted the Dynamic Time Warping (DTW) technique, which is widely rec-

ognized for its effectiveness in addressing temporal distortions and shifts in time series data across various applications. It is important to note that a lower DTW score indicates greater similarity between the two time-series sequences, meaning that they align more closely or exhibit fewer distortions. We computed the DTW distance between trajectories generated by these networks and the true trajectories corresponding to the provided initial positions. Thus, a model with a smaller DTW implies a superior resemblance between network-generated trajectories and the true trajectories. We computed DTW distances between trajectories generated by the networks and the true trajectories corresponding to the provided initial positions. Thus, a model exhibiting a smaller DTW score implies superior performance in extracting fundamental structure inherent to a dataset.

References

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